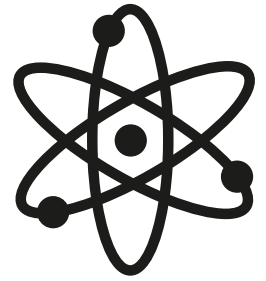


Ingenious inventors

The background to how inventors came up with their ingenious ideas is often fascinating and unexpected. Here are the stories of some very well-known inventors, alongside others whose names are less familiar—though you may well know their inventions!



ANDERSON, MARY (1866–1953)

In 1903, a snowy journey on a streetcar (tram) inspired American inventor Anderson to sketch out an idea for a windshield wiper. At the time, drivers had to lean out of the window and clear their windshields by hand. Her device, made from wood with rubber blades, was controlled by a handle inside the vehicle. Anderson never found a backer for her invention, but in later decades others copied the idea and wipers became a standard feature on most vehicles.

BABBAGE, CHARLES (1791–1871)

This English mathematician and mechanical engineer is often called the “father of the computer.” Babbage worked on various prototypes of his mathematical machines, including the Difference Engine of 1821, used for compiling mathematical tables, and later, the Analytical Engine, which even had a form of memory. His machines were not completed

in his lifetime, but his contribution to computer science is never forgotten.

BERNERS-LEE, TIM (1955–)

Berners-Lee is recognized as the inventor of the World Wide Web. The British software engineer and computer scientist first came up with the idea of people around the world sharing information in 1980, and by 1989 he had developed the global information system, or Web. As director of the Web standards agency, the World Wide Web Foundation, Berners-Lee says: “It has taken all of us to build the Web we have, and now it is up to all of us to build the Web we want—for everyone.”

BRUNEL, ISAMBARD KINGDOM (1806–1859)

Brunel is credited as one of the most significant Britons of all time for his contribution to civil and mechanical engineering. The network of tracks and bridges he built for the Great Western Railway and pioneering work on steamships such as the SS *Great Britain*, launched in 1843 (see p.106), are testament to his genius.

COOPER, MARTIN (1928–)

The first public demonstration of a portable cellular phone, or mobile, was made by the American engineer Martin Cooper in April 1973. At that time, Cooper was employed by Motorola and the call was made on

the Dyna-Tac, a small handheld device on which Cooper and his team had worked. Cooper's other inventions include the first radio-controlled traffic light system and handheld police radios.

DA VINCI, LEONARDO (1452–1519)

The Italian painter and sculptor is best known for his paintings “Mona Lisa” and “The Last Supper.” But he also had a passionate interest in science, architecture, mathematics, engineering, and human anatomy, filling notebooks with sketches of inventions way ahead of the times. He wrote most of his accompanying notes backward, in mirror writing. A flying machine, armored vehicle, helicopter, parachute, and SCUBA gear are among his many designs.

FLEMING, SANDFORD (1827–1915)

In 1876, the Scottish Canadian railway engineer was traveling through Ireland when he missed his train. The timetable, or Fleming's own timing, may have been wrong but it made him consider the need for standardized time. So, he came up with the concept of worldwide standard time. In 1879, Fleming proposed dividing the world into 24 time zones and, in 1884, clocks were reset in line with his ideas.

